By hand calculations

4 december 2024

States

The state space is defined as:

$$O = \{(C,C)\}, \quad 1 = \{(D,C),(C,D),(D,D)\}$$

Payoff Matrix

The payoff matrix for the game is:

	C	D
C	$R = 0.8$	$S = 0$
D	$T = 1$	$P = 0.2$

Q-value Equations

For 0:

1. Cooperation in 0:

$$Q_{00} = (1 - \frac{\varepsilon}{2})\Big[R + \delta \max\{Q_{00}, Q_{01}\}\Big] + \frac{\varepsilon}{2}\Big[S + \delta \max\{Q_{10}, Q_{11}\}\Big]$$

2. Defection in 0:

$$Q_{01} = (1 - \frac{\varepsilon}{2})\Big[T + \delta \max\{Q_{10}, Q_{11}\}\Big] + \frac{\varepsilon}{2}\Big[P + \delta \max\{Q_{10}, Q_{11}\}\Big]$$

For 1:

1. Cooperation in 1:

$$Q_{10} = (1 - \frac{\varepsilon}{2})\Big[S + \delta \max\{Q_{10}, Q_{11}\}\Big] + \frac{\varepsilon}{2}\Big[R + \delta \max\{Q_{00}, Q_{01}\}\Big]$$

2. Defection in 1:

$$Q_{11} = (1 - \frac{\varepsilon}{2})\Big[P + \delta \max\{Q_{10}, Q_{11}\}\Big] + \frac{\varepsilon}{2}\Big[T + \delta \max\{Q_{10}, Q_{11}\}\Big]$$

1 Scenarios

I will define the theoretical Q values of different strategies always against $\varepsilon - GT$

1.1 Case 1: Playing $\varepsilon - GT$

If the opponent plays $\varepsilon - GT$ and I play $\varepsilon - GT$, the Q-values are updated as:

$$Q_{00} = (1-\frac{\varepsilon}{2})\Big[R+\delta Q_{00}\Big] + \frac{\varepsilon}{2}\Big[S+\delta Q_{11}\Big]$$
$$Q_{01} = (1-\frac{\varepsilon}{2})\Big[T+\delta Q_{11}\Big] + \frac{\varepsilon}{2}\Big[P+\delta Q_{11}\Big]$$
$$Q_{10} = (1-\frac{\varepsilon}{2})\Big[S+\delta Q_{11}\Big] + \frac{\varepsilon}{2}\Big[R+\delta Q_{00}\Big]$$
$$Q_{11} = (1-\frac{\varepsilon}{2})\Big[P+\delta Q_{11}\Big] + \frac{\varepsilon}{2}\Big[T+\delta Q_{11}\Big]$$

1.1.1 Solving this system of equations

The equations for Q_{ij} are derived as follows:

Equation for Q_{00}:

$$Q_{00} = (1-\frac{\varepsilon}{2})(R+\delta Q_{00}) + \frac{\varepsilon}{2}(S+\delta Q_{11})$$
$$Q_{00} = (1-\frac{\varepsilon}{2})R + (1-\frac{\varepsilon}{2})\delta Q_{00} + \frac{\varepsilon}{2}S + \frac{\varepsilon}{2}\delta Q_{11}$$
$$Q_{00} - (1-\frac{\varepsilon}{2})\delta Q_{00} = (1-\frac{\varepsilon}{2})R + \frac{\varepsilon}{2}S + \frac{\varepsilon}{2}\delta Q_{11}$$
$$Q_{00}\Big[1-(1-\frac{\varepsilon}{2})\delta\Big] = (1-\frac{\varepsilon}{2})R + \frac{\varepsilon}{2}S + \frac{\varepsilon}{2}\delta Q_{11}$$
$$Q_{00} = \frac{(1-\frac{\varepsilon}{2})R + \frac{\varepsilon}{2}S + \frac{\varepsilon}{2}\delta Q_{11}}{1-(1-\frac{\varepsilon}{2})\delta}$$

Equation for Q_{01}:

$$Q_{01} = (1-\frac{\varepsilon}{2})(T+\delta Q_{11}) + \frac{\varepsilon}{2}(P+\delta Q_{11})$$
$$Q_{01} = (1-\frac{\varepsilon}{2})T + (1-\frac{\varepsilon}{2})\delta Q_{11} + \frac{\varepsilon}{2}P + \frac{\varepsilon}{2}\delta Q_{11}$$
$$Q_{01} = (1-\frac{\varepsilon}{2})T + \frac{\varepsilon}{2}P + \delta Q_{11}$$

Equation for Q_{10}:

$$Q_{10} = (1-\frac{\varepsilon}{2})(S+\delta Q_{11}) + \frac{\varepsilon}{2}(R+\delta Q_{00})$$
$$Q_{10} = (1-\frac{\varepsilon}{2})S + (1-\frac{\varepsilon}{2})\delta Q_{11} + \frac{\varepsilon}{2}R + \frac{\varepsilon}{2}\delta Q_{00}$$

Equation for Q_{11}:

$$\begin{aligned}
Q_{11} &= (1-\frac{\varepsilon}{2})\,(P+\delta Q_{11}) + \frac{\varepsilon}{2}\,(T+\delta Q_{11}) \\
Q_{11} &= (1-\frac{\varepsilon}{2})P + (1-\frac{\varepsilon}{2})\delta Q_{11} + \frac{\varepsilon}{2}T + \frac{\varepsilon}{2}\delta Q_{11} \\
Q_{11} - [(1-\frac{\varepsilon}{2})\delta + \frac{\varepsilon}{2}\delta]Q_{11} &= (1-\frac{\varepsilon}{2})P + \frac{\varepsilon}{2}T \\
Q_{11}\,[1-\delta] &= (1-\frac{\varepsilon}{2})P + \frac{\varepsilon}{2}T \\
Q_{11} &= \frac{(1-\frac{\varepsilon}{2})P + \frac{\varepsilon}{2}T}{1-\delta}
\end{aligned}$$

Final System of Equations:

$$\begin{aligned}
Q_{00} &= \frac{(1-\frac{\varepsilon}{2})R + \frac{\varepsilon}{2}S + \frac{\varepsilon}{2}\delta Q_{11}}{1-(1-\frac{\varepsilon}{2})\delta} \\
Q_{01} &= (1-\frac{\varepsilon}{2})T + \frac{\varepsilon}{2}P + \delta Q_{11} \\
Q_{10} &= (1-\frac{\varepsilon}{2})S + (1-\frac{\varepsilon}{2})\delta Q_{11} + \frac{\varepsilon}{2}R + \frac{\varepsilon}{2}\delta Q_{00} \\
Q_{11} &= \frac{(1-\frac{\varepsilon}{2})P + \frac{\varepsilon}{2}T}{1-\delta}
\end{aligned}$$

Substituting one in another: The reduced system of equations is given as follows:

$$Q_{00} = \frac{(1-\frac{\varepsilon}{2})R + \frac{\varepsilon}{2}S + \frac{\varepsilon}{2}\delta \cdot \frac{(1-\frac{\varepsilon}{2})P + \frac{\varepsilon}{2}T}{1-\delta}}{1-(1-\frac{\varepsilon}{2})\delta}$$

$$Q_{10} = (1-\frac{\varepsilon}{2})T + \frac{\varepsilon}{2}P + \delta \cdot \frac{(1-\frac{\varepsilon}{2})P + \frac{\varepsilon}{2}T}{1-\delta}$$

$$Q_{01} = (1-\frac{\varepsilon}{2})S + (1-\frac{\varepsilon}{2})\delta \cdot \frac{(1-\frac{\varepsilon}{2})P + \frac{\varepsilon}{2}T}{1-\delta} + \frac{\varepsilon}{2}R + \frac{\varepsilon}{2}\delta Q_{00}$$

$$Q_{11} = \frac{(1-\frac{\varepsilon}{2})P + \frac{\varepsilon}{2}T}{1-\delta}$$

1.2 Case 2: Playing AD

If the opponent plays $\varepsilon - GT$ and I play AD, the Q-values are updated as :

$$\begin{aligned}
Q_{00} &= (1-\frac{\varepsilon}{2})\Big[R + \delta Q_{01}\Big] + \frac{\varepsilon}{2}\Big[S + \delta Q_{11}\Big] \\
Q_{01} &= (1-\frac{\varepsilon}{2})\Big[T + \delta Q_{11}\Big] + \frac{\varepsilon}{2}\Big[P + \delta Q_{11}\Big] \\
Q_{10} &= (1-\frac{\varepsilon}{2})\Big[S + \delta Q_{11}\Big] + \frac{\varepsilon}{2}\Big[R + \delta Q_{01}\Big] \\
Q_{11} &= (1-\frac{\varepsilon}{2})\Big[P + \delta Q_{11}\Big] + \frac{\varepsilon}{2}\Big[T + \delta Q_{11}\Big]
\end{aligned}$$

The reduced system of equations is given as follows:

$$Q_{00} = (1-\frac{\varepsilon}{2})\Big[R + \delta Q_{01}\Big] + \frac{\varepsilon}{2}\Big[S + \delta Q_{11}\Big]$$

$$Q_{01} = (1-\frac{\varepsilon}{2}) \cdot T + \frac{\varepsilon}{2} \cdot P + \delta \cdot Q_{11}$$
$$Q_{10} = (1-\frac{\varepsilon}{2}) \Big[S + \delta Q_{11} \Big] + \frac{\varepsilon}{2} \Big[R + \delta Q_{01} \Big]$$
$$Q_{11} = \frac{(1-\frac{\varepsilon}{2}) \cdot P + \frac{\varepsilon}{2} \cdot T}{1-\delta}$$

1.3 Case 3: Playing *AC*

$$Q_{00} = (1-\frac{\varepsilon}{2}) \Big[R + \delta Q_{00} \Big] + \frac{\varepsilon}{2} \Big[S + \delta Q_{10} \Big]$$
$$Q_{01} = (1-\frac{\varepsilon}{2}) \Big[T + \delta Q_{10} \Big] + \frac{\varepsilon}{2} \Big[P + \delta Q_{10} \Big]$$
$$Q_{10} = (1-\frac{\varepsilon}{2}) \Big[S + \delta Q_{10} \Big] + \frac{\varepsilon}{2} \Big[R + \delta Q_{00} \Big]$$
$$Q_{11} = (1-\frac{\varepsilon}{2}) \Big[P + \delta Q_{10} \Big] + \frac{\varepsilon}{2} \Big[T + \delta Q_{10} \Big]$$

The reduced system of equations is given as follows:

$$Q_{00} = \frac{(1-\frac{\varepsilon}{2}) \cdot R + \frac{\varepsilon}{2} \cdot S + \frac{\varepsilon}{2} \cdot \delta \cdot Q_{10}}{1-(1-\frac{\varepsilon}{2}) \cdot \delta}$$
$$Q_{01} = (1-\frac{\varepsilon}{2}) \cdot T + \frac{\varepsilon}{2} \cdot S + \frac{\varepsilon}{2} \cdot Q_{10}$$
$$Q_{10} = \frac{(1-\frac{\varepsilon}{2}) \cdot S + \frac{\varepsilon}{2} \cdot R + \frac{\varepsilon}{2} \cdot \delta Q_{00}}{1-(1-\frac{\varepsilon}{2}) \cdot \delta}$$
$$Q_{11} = (1-\frac{\varepsilon}{2}) \cdot P + \frac{\varepsilon}{2} \cdot T + \delta \cdot Q_{10}$$

1.4 Case 4: Playing *AGT*

If the opponent plays $\varepsilon - GT$ and I play AGT, that means defect in (C,C) and cooperate otherwise, the Q-values are updated as:

$$Q_{00} = (1-\frac{\varepsilon}{2}) \Big[R + \delta Q_{01} \Big] + \frac{\varepsilon}{2} \Big[S + \delta Q_{01} \Big]$$
$$Q_{01} = (1-\frac{\varepsilon}{2}) \Big[T + \delta Q_{10} \Big] + \frac{\varepsilon}{2} \Big[P + \delta Q_{10} \Big]$$
$$Q_{10} = (1-\frac{\varepsilon}{2}) \Big[S + \delta Q_{10} \Big] + \frac{\varepsilon}{2} \Big[R + \delta Q_{01} \Big]$$
$$Q_{11} = (1-\frac{\varepsilon}{2}) \Big[P + \delta Q_{10} \Big] + \frac{\varepsilon}{2} \Big[T + \delta Q_{10} \Big]$$

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